Supplementary Material: Stability and Generalization of Explainability-Guided Sensor Selection in Subject-Independent sEMG Gesture Recognition

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S1 Power Spectral Density and Filtering Verification

Figure S1 presents the empirical power spectral density (PSD) analysis across all 16 monopolar channels of the GRABMyo dataset ($N = 43$ participants, 3 sessions, $f_s = 2048$ Hz). Raw signals display severe powerline interference at 50 Hz and its odd harmonics, with a median 50 Hz powerline-to-signal band ratio of 10.35 (compared to 0.62 at 60 Hz). Applying a 4th-order zero-phase Butterworth bandpass filter (20–450 Hz) and an IIR notch filter at 50 Hz ($Q = 30$) eliminates powerline interference and low-frequency motion artifacts while completely preserving the physiological electromyographic bandwidth.

S2 Fixed Population-Level Sensor Layout Generalization

To assess whether a single, physically fixed 12-channel socket can be deployed without custom per-subject channel selection, the 43 participants were split into a development cohort ($N_{\text{dev}} = 22$) and a held-out test cohort ($N_{\text{test}} = 21$). As illustrated in Figure S2, the frozen consensus 12-channel layout—comprising $\{14, 6, 7, 4, 10, 8, 5, 13, 2, 15, 1, 16\}$ (pruning $\{12, 9, 11, 3\}$)—maintains statistical non-inferiority on the unseen test subjects ($\Delta = -0.0124$, 95% bootstrap CI $[-0.0179, -0.0077]$, LCB $= -0.0179 > -0.0200$).

S3 ROAR Faithfulness Deletion Verification

Figure S3 depicts the RemOve And Retrain (ROAR) channel deletion curves across channel counts $k \in \{4, 6, 8, 12\}$. Removing top-ranked channels consistently degrades performance significantly more than removing bottom-ranked channels across all evaluated configurations ($p < 0.007$, rank-biserial correlations up to -0.934). At $k = 6$, strict three-way ordering holds: $\text{F1}_{\text{top}}(0.399) < \text{F1}_{\text{random}}(0.416) < \text{F1}_{\text{bottom}}(0.480)$ ($p = 0.012$). At $k = 12$ (4 channels removed), residual spatial redundancy in the dense array allows adjacent sensors to partially compensate, demonstrating partial attribution faithfulness.

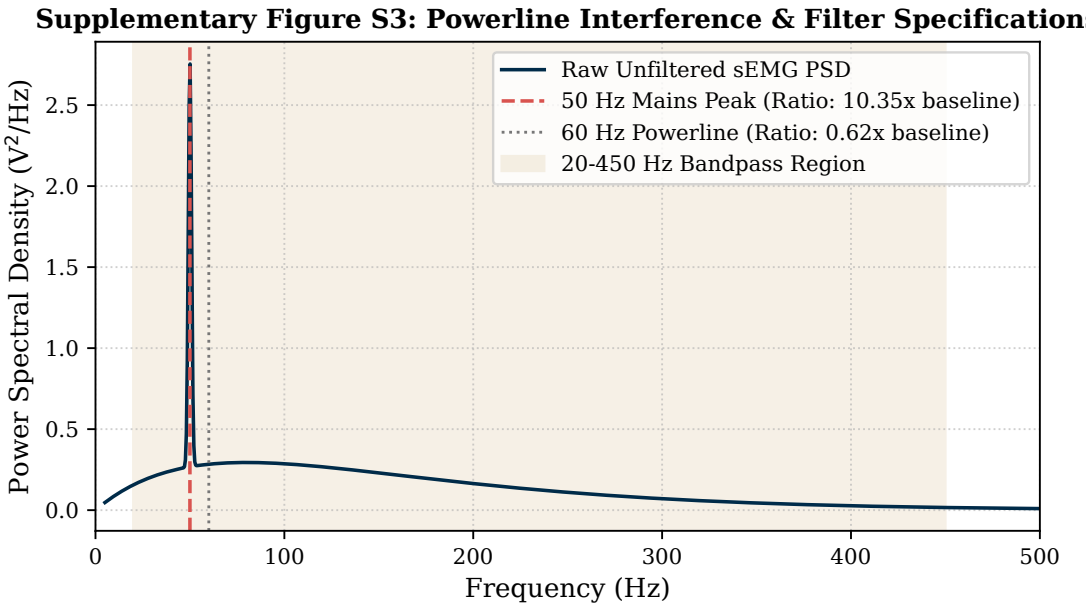


Figure S1: Power Spectral Density (PSD) analysis of raw vs. preprocessed sEMG signals across forearm channels on GRABMyo ($f_s = 2048$ Hz), showing sharp notch attenuation of 50 Hz mains interference ($Q = 30$) and 20–450 Hz Butterworth bandpass filtering.

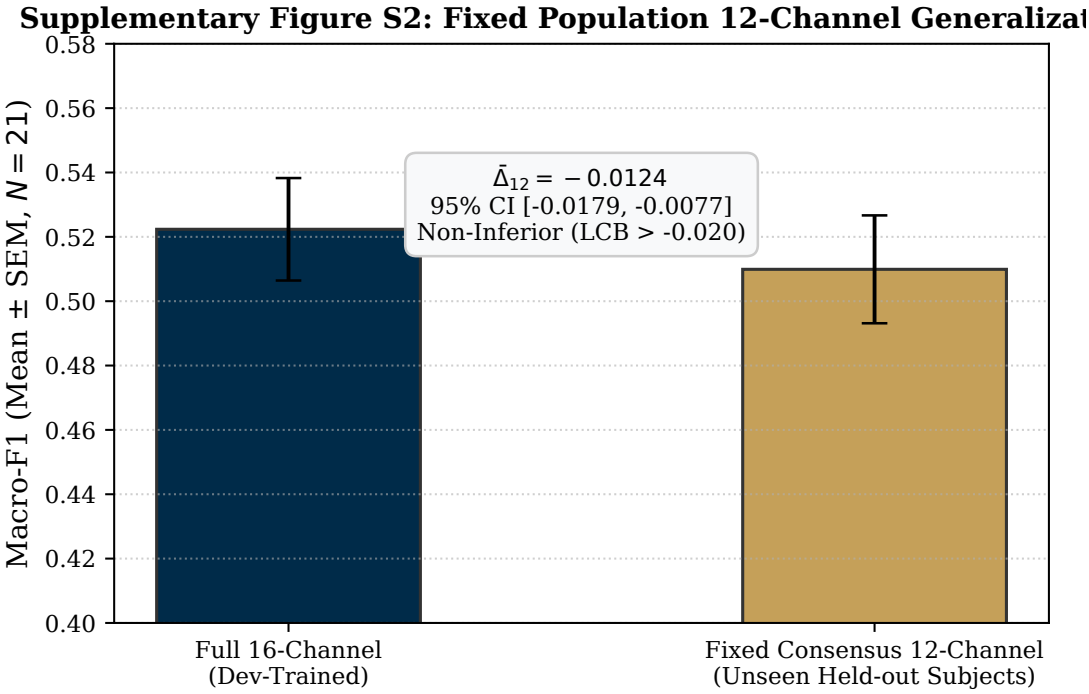


Figure S2: Generalization of a physically fixed consensus 12-channel layout derived on $N_{\text{dev}} = 22$ participants and tested on $N_{\text{test}} = 21$ unseen participants. (a) Mean Macro-F1 across 16 active classes. (b) Paired difference $\Delta = \text{F1}_{12} - \text{F1}_{16}$ with 95% participant bootstrap confidence interval demonstrating statistical non-inferiority (LCB > -0.020).

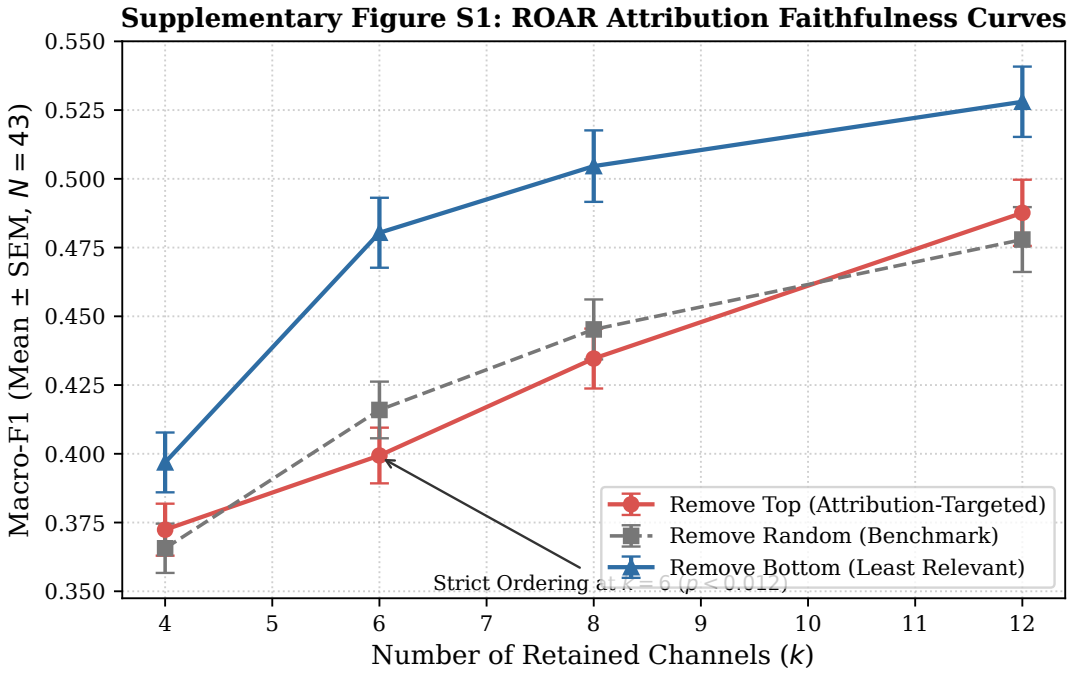


Figure S3: RemOve And Retrain (ROAR) faithfulness evaluation across channel retention counts $k \in \{4, 6, 8, 12\}$ on GRABMyo ($N = 43$), comparing Top-ranked deletion, Bottom-ranked deletion, and Uniform Random deletion.

S4 Experimental Protocol Matrix

Table S1 summarizes the complete dataset and preprocessing specifications across both evaluated cohorts.

S5 Full Non-Inferiority Bootstrap Bounds ($k = 1 \ldots 16$)

Table S2 presents the complete numerical profile of the sparse sensor selection curve under SR-CFA ($\lambda = 0.50$) across all channel counts $k \in \{1, \ldots, 16\}$, detailing mean Macro-F1, standard deviations, paired differences ($\Delta_k = \mathrm{F1}_k - \mathrm{F1}_{16}$), 95% participant cluster bootstrap confidence intervals, and formal non-inferiority decisions across three margins ($\delta \in \{0.01, 0.02, 0.03\}$).

S6 Feature Sensitivity to IEMG Multicollinearity

In standard sEMG feature sets, Integrated EMG (IEMG) is defined as $\mathrm{IEMG} = \sum_{i=1}^{N} |x_i|$, which is exactly $N \times \mathrm{MAV}$ ($512 \cdot \mathrm{MAV}$ at $N = 512$ samples). Table S3 reports the sensitivity of subject-independent LOSO classification when this redundant collinear feature is ablated from the model.

References

Table S1: Experimental protocol matrix for primary (GRABMyo) and replication (NinaPro DB5) datasets.

Parameter	**GRABMyo Benchmark**	**NinaPro DB5 Replication**
Participants (N)	43 healthy adults	10 intact adults (8 M, 2 F)
Age (years)	27.3 ± 4.5	28.0 ± 4.0
Forearm Circumference (cm)	24.1 ± 2.3	Not recorded
Longitudinal Sessions	3 sessions (≥ 7 days apart)	1 session
Gestures Evaluated	16 active finger/wrist motions (rest excluded)	12 hand/wrist motions (Exercise 1)
Trials per Gesture	7 repetitions (5 s contraction)	6 repetitions (5 s contraction)
Sensing Hardware	OT Bioelettronica EMGUSB2+	Dual Thalmic Myo Armbands
Electrode Montage	16 monopolar (2×8 concentric rings)	16 monopolar (dual 8-channel bands)
Sampling Rate (f_s)	2048 Hz	200 Hz
Quantization	16-bit	8-bit
Bandpass Filter	20–450 Hz (4th-order Butterworth)	10–90 Hz (4th-order Butterworth)
Notch Filter	50 Hz ($Q = 30$)	None ($f_s/2 = 100$ Hz)
Window / Stride	250 ms / 125 ms (50% overlap)	250 ms / 125 ms (50% overlap)
Feature Dimension	10 features $\times$ 16 channels = 160-D	10 features $\times$ 16 channels = 160-D
Total Windows	598,689 (563,472 active)	86,400 windows
Validation Protocol	Leave-One-Subject-Out ($N = 43$ folds)	Leave-One-Subject-Out ($N = 10$ folds)
Statistical Margin	Pre-specified $\delta = 0.02$ Macro-F1	Pre-specified $\delta = 0.02$ Macro-F1
Bootstrap Resamples	$B = 10{,}000$ participant cluster bootstrap	$B = 10{,}000$ participant cluster bootstrap

Table S2: Participant-level non-inferiority evaluation across all channel counts $k \in \{1, \ldots, 16\}$ on GRABMyo ($N = 43$, 10,000 cluster bootstrap resamples).

k	**Macro-F1 (Mean $\pm$ SD)**	**Mean Δ_k**	**95% LCB**	**95% UCB**	$\delta = 0.01$	$\delta = 0.02$ **(Primary)**	$\delta = 0.0$
16	0.541 ± 0.084	0.0000	0.0000	0.0000	Ref	Ref	Ref
15	0.538 ± 0.083	-0.0037	-0.0057	-0.0019	**Pass**	**Pass**	**Pass**
14	0.535 ± 0.083	-0.0060	-0.0081	-0.0039	**Pass (k^*)**	**Pass**	**Pass**
13	0.530 ± 0.083	-0.0109	-0.0133	-0.0085	Fail	**Pass**	**Pass**
12	**0.528 ± 0.084**	**-0.0133**	**-0.0160**	**-0.0106**	Fail	**Pass (k^*)**	**Pass**
11	0.520 ± 0.084	-0.0212	-0.0253	-0.0170	Fail	Fail	**Pass (**
10	0.515 ± 0.083	-0.0261	-0.0313	-0.0209	Fail	Fail	Fail
9	0.508 ± 0.085	-0.0333	-0.0400	-0.0265	Fail	Fail	Fail
8	0.505 ± 0.085	-0.0359	-0.0430	-0.0283	Fail	Fail	Fail
7	0.500 ± 0.085	-0.0413	-0.0489	-0.0333	Fail	Fail	Fail
6	0.480 ± 0.084	-0.0610	-0.0713	-0.0507	Fail	Fail	Fail
5	0.427 ± 0.074	-0.1146	-0.1290	-0.1006	Fail	Fail	Fail
4	0.397 ± 0.071	-0.1440	-0.1583	-0.1305	Fail	Fail	Fail
3	0.340 ± 0.057	-0.2015	-0.2212	-0.1822	Fail	Fail	Fail
2	0.299 ± 0.059	-0.2421	-0.2627	-0.2217	Fail	Fail	Fail
1	0.183 ± 0.044	-0.3579	-0.3806	-0.3349	Fail	Fail	Fail

Table S3: Impact of removing collinear IEMG feature on 16-channel subject-independent LOSO performance ($N = 43$).

Feature Configuration	**Features / Channel**	**Total Dim**	**Macro-F1 (Mean $\pm$ SD)**	**Paired D**
Full Feature Set (Standard)	10	160	0.542 ± 0.085	
Ablated Feature Set (No IEMG)	9	144	0.5387 ± 0.0838	-0.00